

CSA1014 SOFTWARE ENGINEERING

CO-1 ASSESSMENT TOOL - 1

Warehouse Robotics Coordination and Inventory Automation Platform

BY

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Problem Statement & Scope

Operational Bottlenecks

- **Rapid Fulfillment Demands:** Modern e-commerce mandates sub-24-hour processing, causing severe operational pressure.
- **Multi-Robot Gridlocks:** Uncoordinated Autonomous Mobile Robots (AMRs) experience collisions and deadlock at narrow intersections.
- **Inventory Desynchronization:** Manual logging creates lag between physical shelves and warehouse management databases.



Stakeholder Requirements



Warehouse Operators

Goal: Live fleet monitoring and exception handling.

Challenge: UI fatigue during peak operating hours.

Solution: Single-click override controls and streamlined telemetry UI.



System Engineers

Goal: Maintain uptime and API stability.

Challenge: Wi-Fi latency spikes and multi-brand protocols.

Solution: Standard ROS2 adapters and modular REST APIs.



Executive Leadership

Goal: Maximize order throughput and return on investment.

Challenge: High upfront capital costs and downtime risks.

Solution: Proven ROI metrics and non-disruptive cloud scaling.

Legacy Systems vs Solution

Evaluation Dimension	Legacy / Manual Systems	Proposed Automation Platform
Motion Planning	Static routes; frequent aisle congestion	Dynamic multi-agent routing (A* / CBS algorithms)
Inventory Updates	Batch barcode scanning; periodic lag	Real-time automated RFID / Vision updates (<100ms)
Fleet Scalability	Fails past 50 active robotic agents	Horizontally scales to 200+ agents via Kubernetes
Hardware Compatibility	Locked to single robot manufacturer	Vendor-agnostic microservice protocol adapters

Architecture & SE Principles

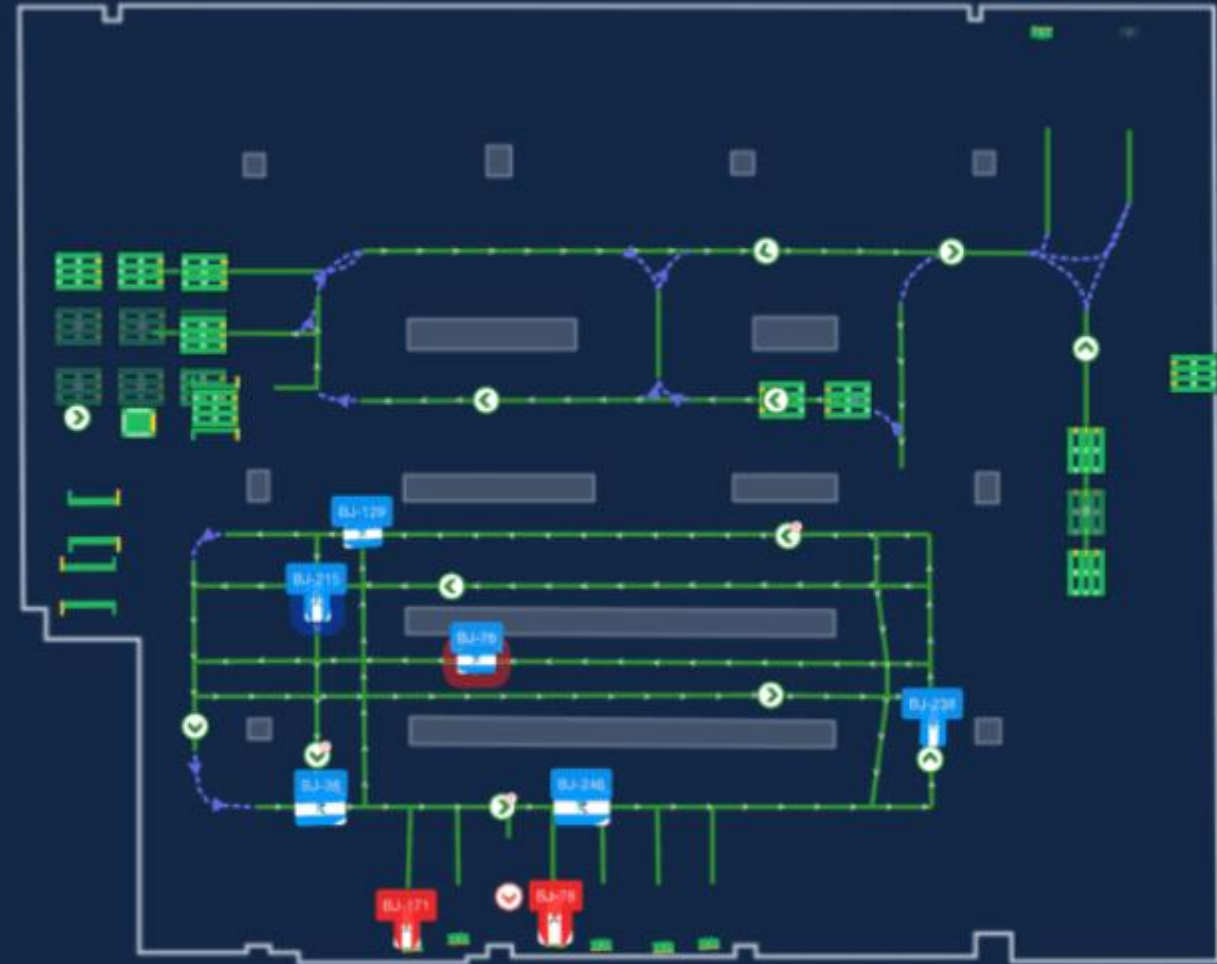
Microservices & Engineering Standards

Modular Stack: Independent services for Fleet Routing, Inventory State, and Analytics using Node.js, Go, and Apache Kafka.

Scalability & Resilience: Dockerized containers managed by Kubernetes ensure auto-scaling during surge hours.

Security Protocols: End-to-end TLS encryption and Role-Based Access Control (RBAC) safeguard machinery control channels.

Performance Hub



Agile & Simulation Strategy

Agile Scrum Methodology

2-Week Sprints: Enables rapid feature delivery, from pathfinding algorithms to operator UI controls.

Risk Mitigation: Physical integration risks are mitigated via continuous feedback loops and weekly stakeholder reviews.

Simulation Verification

ROS2 / Gazebo Digital Twin: Safely tests collision avoidance in a virtual 3D environment before deployment.

Hardware-in-the-Loop: Simulates Wi-Fi signal drops and stress-tests system limits with 500+ active agents.

Expected Metrics & Outcomes

+40%

Throughput Improvement

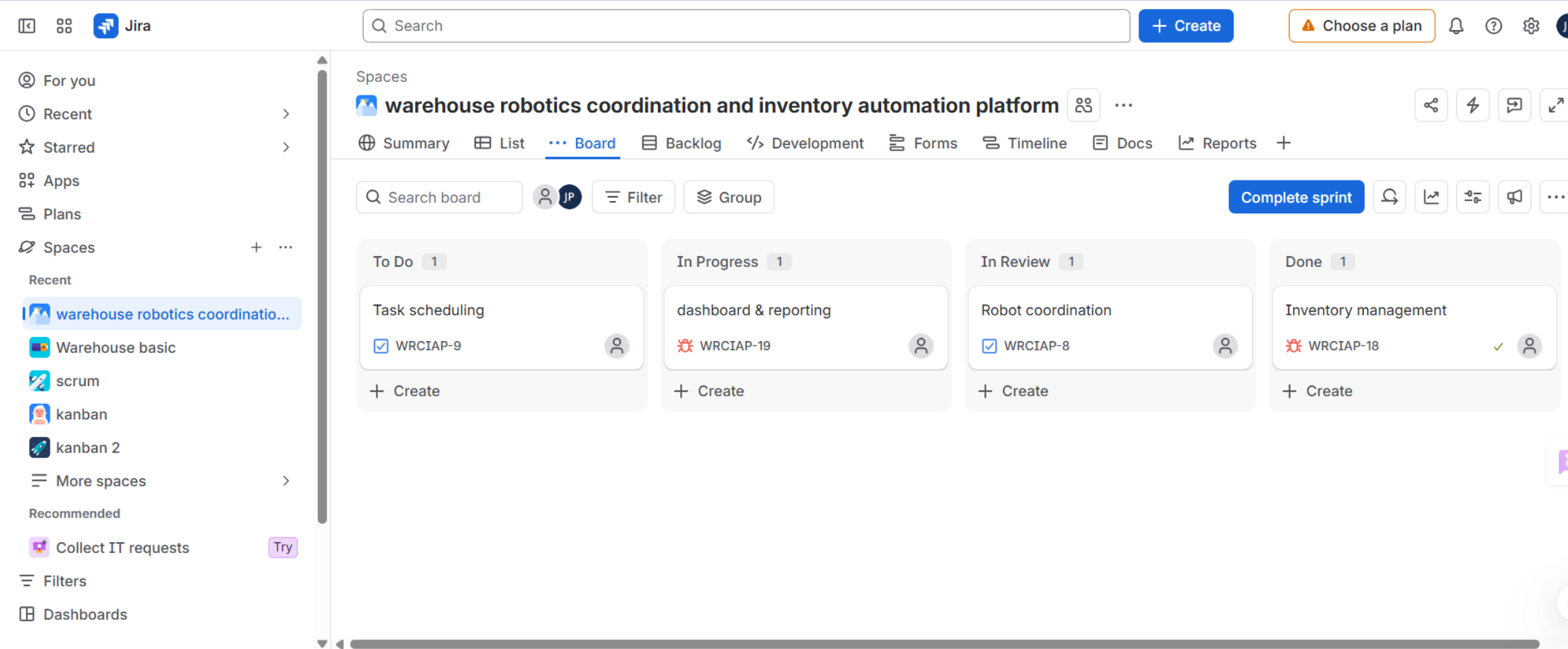
Key Benefits & Limitations

Quantitative Impact: Fulfills orders 35% faster, maintains 99.9% inventory tracking precision, and eliminates physical collisions.

System Constraints: Requires high-density Enterprise Wi-Fi 6 or Private 5G infrastructure and initial 3D grid mapping.

Future Scope: Integration of AI predictive maintenance for robot hardware and 5G edge compute offloading.

Jira Implementation



THANK YOU

Thank you for your time and attention!

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